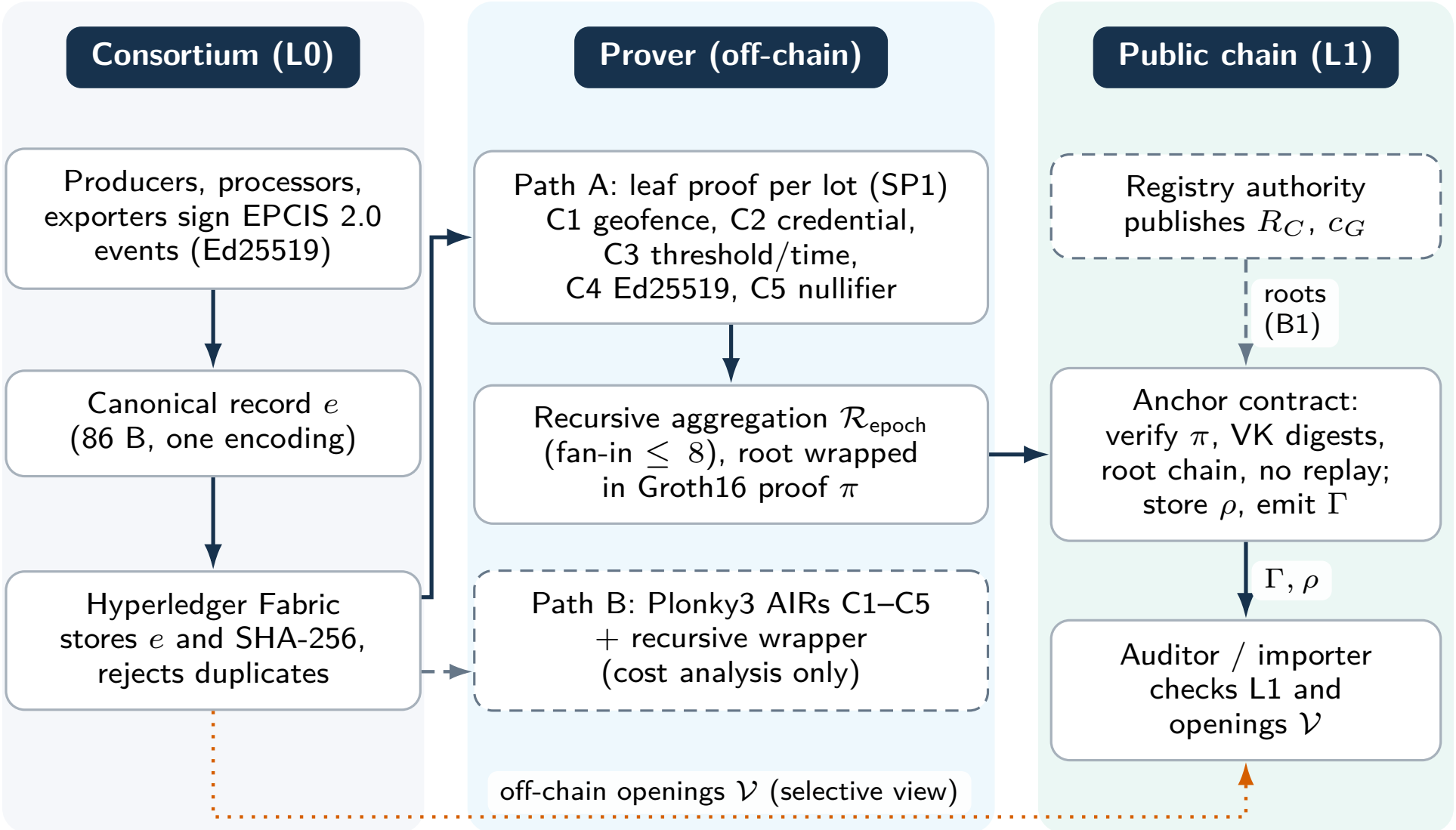

Consortium (L0)
Prover (off-chain)
Public chain (L1)
Producers, processors, exporters sign EPCIS 2.0 events (Ed25519)
Canonical record e (86 B, one encoding)
Hyperledger Fabric stores e and SHA-256, rejects duplicates
Path A: leaf proof per lot (SP1) C1 geofence, C2 credential, C3 threshold/time, C4 Ed25519, C5 nullifier
Recursive aggregation $\mathcal{R}_{\text{epoch}}$ (fan-in $\leq$ 8), root wrapped in Groth16 proof π
Path B: Plonky3 AIRs C1–C5 + recursive wrapper (cost analysis only)
Registry authority publishes R_C, c_G
roots (B1)
Anchor contract: verify π, VK digests, root chain, no replay; store ρ, emit Γ
Γ, ρ
Auditor / importer checks L1 and openings $\mathcal{V}$
off-chain openings $\mathcal{V}$ (selective view)